

Can You Feel It?

Cem Baykal, 1878560

Abstract: In this project, artificial intelligence is created to predict humans' current feelings by analyzing their faces' current shape. Using these analysis results and knowing humans' feelings, better interactivity can be obtained between AI and the people. Passive learning is used to initialize artificial intelligence. Then, to be more functional, active learning is used, which means AI can decide whether the upcoming data is new/old.

I. Introduction

No matter what, AI will become a massive part of our lives, and the predicted final goal is to make AI a perfect servant to humans. To better understand humans, AI has to catch humans' feelings, and in this project, an emotion detector is created.

II. How to detect emotion?

Kaggle FER2013 Dataset, which contains 48 by 48 grayscale face images, is used as training data set, and it is shared in the Github repository. The artificial neural network is created with CNN and fully connected neural networks. CNN is used as a feature extractor with 2048 features, and then the fully connected neural network is used to differentiate seven different labels.

```
emotion_model = Sequential()
emotion_model.add(Conv2D(32, kernel_size=(3, 3), activation='relu', input_shape=(48, 48, 1)))
emotion_model.add(Conv2D(64, kernel_size=(3, 3), activation='relu'))
emotion_model.add(MaxPooling2D(pool_size=(2, 2)))
emotion_model.add(Dropout(0.25))
emotion_model.add(Conv2D(128, kernel_size=(3, 3), activation='relu'))
emotion_model.add(MaxPooling2D(pool_size=(2, 2)))
emotion_model.add(Conv2D(128, kernel_size=(3, 3), activation='relu'))
emotion_model.add(MaxPooling2D(pool_size=(2, 2)))
emotion_model.add(Dropout(0.25))

emotion_model.add(Flatten())
emotion_model.add(Dense(4096, activation='relu'))
emotion_model.add(Dropout(0.2))
emotion_model.add(Dense(2048, activation='relu'))
emotion_model.add(Dropout(0.1))
emotion_model.add(Dense(512, activation='relu'))
emotion_model.add(Dense(7, activation='softmax'))
emotion_model.summary()
```

Figure 1 - The Neural Network Architecture

The training result is 62% accuracy, which is not perfect; but, it is a good start. After the training is completed, feature vectors of all images are generated to obtain the feature matrix.

Two different approaches are analyzed with the feature matrix, encoders, and Singular Value Decomposition. With the encoders and decoders, the feature vector's size decreased to two or three to visually analyze the data. However, it failed.

SVD is applied to this feature matrix, and singular values are shown in Figure 2. Even though it seems only the first eigenvalue is the important one, only 20% of the total energy is in the first eigenvalue. Gavish and Donoho[1] suggestion decreases the feature vector size to 430 by keeping 80% of the energy. SVD indicates total energy is stored in 1778 features. Due to time limitations, I do not try to decrease the feature vector's size, and I will use it as it is, with 2048 features.

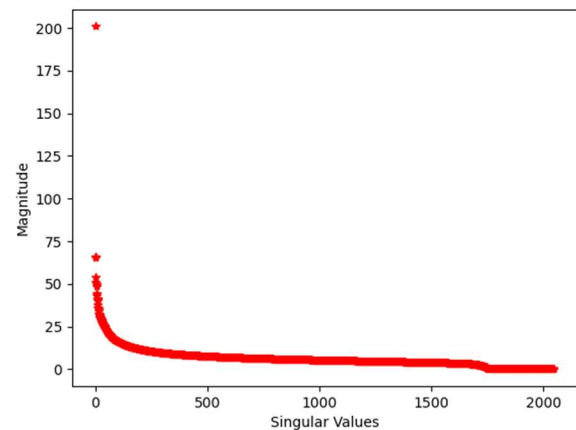


Figure 2 - Singular values of the feature matrix.

III. Making the final decision

Making decisions relies on the following procedure.

1. Predict with the model with $n \times 1$ output prediction vector, where n represents the number of emotions trained.
2. Sort the vector in decreasing order.
3. Check if $P_0(0)_{n \times 1} > 2P_0(1)_{n \times 1}$.
 - a. True. AI is sure that it makes the right decision.
 - b. False. AI is not sure. The answer is unknown. It may be something new. Capture the input and store it in the "TBD" folder.

Whenever the size of the "TBD" folder is bigger than 20:

1. Ask the user about the new data. What is that emotion? Get the keyword.
2. Go to google.com or any other search engine. Download some images with the given keyword. Preprocess the downloaded files.
3. Copy the AI architecture and change the output vector dimension to $P_{1(n+1) \times 1}$.
4. Randomly select images from the training set and the TBD folder. Retrain the neural network and check the accuracy if it is acceptable.
 - a. True. Congratulations, you have learned a new thing.
 - b. False. Discard the second list.

The procedure is also shown in the flowchart at the GitHub repository and the end of this report.

IV. Results

CascadeClassifier() from the OpenCV is used to catch the human faces, and some examples are given in Figure 3.

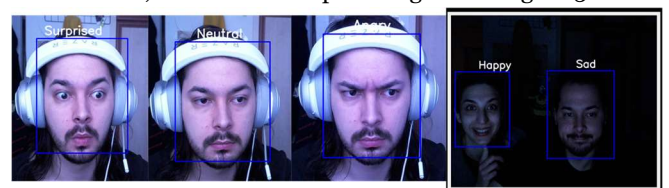


Figure 3 - Example results

V. Conclusion and future works

In this project, artificial intelligence was created to detect humans emotions from their facial expressions. This part may be implemented into a robot and make it catch humans' feelings. Before implemented it into a robot, it may be better to make the following changes.

60% accuracy is not enough to continue this project, and as future work, a different dataset should be used. Secondly, CascadeClassifier() is not a good face catcher, especially in the rotated cases. Other algorithms may provide better performances. Learning the new algorithm is not working since it is not implemented yet; however, it can capture the emotions that are not trained and label them as unknown. Some cases related to these problems are depicted in Figure 4.

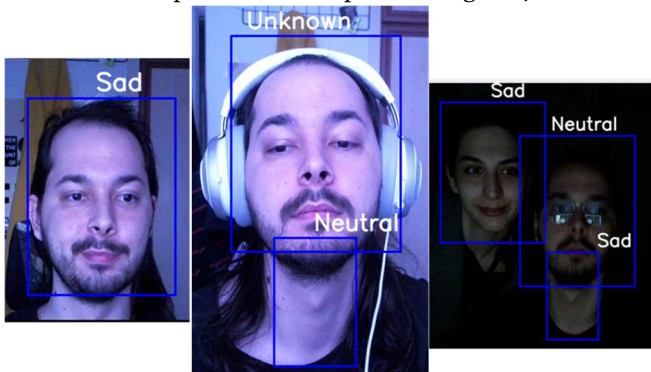


Figure 4 – Example fallbacks.

References

[1] M. Gavish and D. L. Donoho, "The Optimal Hard Threshold for Singular Values is $4/\sqrt{3}$," in IEEE Transactions on Information Theory, vol. 60, no. 8, pp. 5040-5053, Aug. 2014, doi: 10.1109/TIT.2014.2323359.

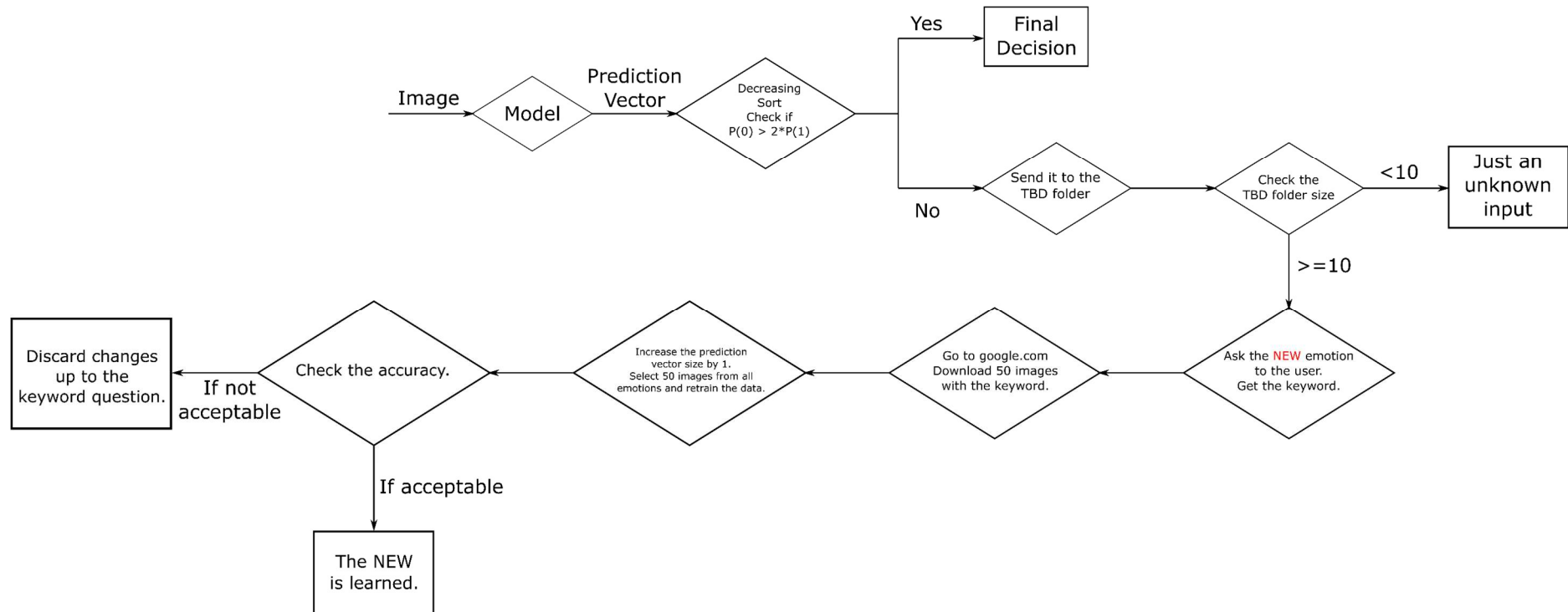


Figure 5 - The Flow Chart to make a decision.